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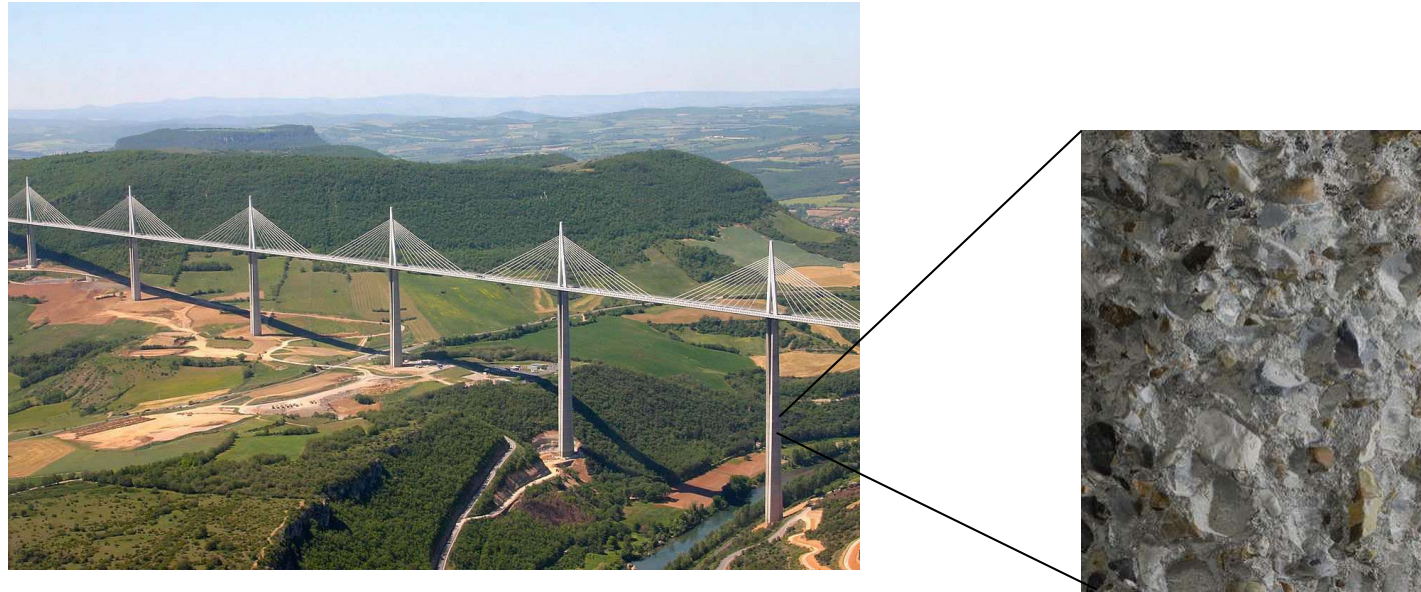
Stochastic Finite Element Methods

#3: Random Fields

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Introduction

A random field $H(x, \theta)$, where $\theta \in \Theta$ is associated to the probability space $(\Theta, \mathcal{F}, \mathcal{P})$, is a continuous field. It describes the spatial variability of a random variable.



Within the context of FEM, $H(x, \theta)$ has to be assigned to the spatial FEM discretization of the domain $\mathcal{D}$. For convenience, it is discretized

$$H(x, \theta) \simeq \hat{H}(x_N, \theta), \quad \text{with} \quad x_N = \sum_n^N x_n$$

Representation of Random Fields

- ❑ Point discretization methods: Each random variable X_i is a selected value of the random field $H(\cdot)$ at some given point x_i
- ❑ Average discretization methods: the set of random variables $\{X_i\}$ are weighted integrals of $H(\cdot)$ over a domain Ω_e

$$X_i = \int_{\Omega_e} H(x) w(x) \, d\Omega_e$$

with the weighting function $w(x)$.

- ❑ Series expansion methods

Gaussian Fields

Analytical form for simple geometries.

- Established theories behind
- Explicit formulas to predict and to compare with experiments

Gaussian fields are fully described by its mean and covariance function.

Point Discretization Methods

- ❑ Midpoint method: A RV is assigned to the center-points of the elements
- ❑ Shape-function method: A RV is assigned to the nodal points of the elements and interpolated by shape-functions
- ❑ Integration point method: A RV is assigned to each integration point
- ❑ Optimal Linear Estimation (OLE) Method (also known as Kriging)

$$\hat{H}(\mathbf{x}) = a(\mathbf{x}) + \sum_{i=1}^{n_a} b_i(\mathbf{x}) X_i = a(\mathbf{x}) + \mathbf{b}^T(\mathbf{x}) \cdot \mathbf{X}, \quad \mathbf{x} \in \Omega$$

where $a(x)$ and $b_i(x)$ are computed by minimizing the variance error.

Averaging Discretization Methods

- Spatial Averaging method:

$$\hat{H}(\mathbf{x}) = \frac{\int_{\Omega_e} H(\mathbf{x}) d\Omega_e}{|\Omega_e|} = \bar{H}_e, \quad \mathbf{x} \in \Omega_e$$

- Weighted Integral method: $\mathbb{C}(\mathbf{x}, \theta) = \mathbb{C}_0 [1 + H(\mathbf{x}, \theta)]$

the element stiffness matrix reads

$$\mathbf{k}^e(\theta) = \mathbf{k}_0^e + \Delta \mathbf{k}^e(\theta) \quad \text{with} \quad \Delta \mathbf{k}^e(\theta) = \int_{\Omega_e} H(\mathbf{x}, \theta) \mathbf{B}^T \mathbb{C}_0 \mathbf{B} d\Omega_e$$

where

$$\mathbf{k}^e(\theta) = \mathbf{k}_0^e + \sum_{i=1}^{n_{WI}} \Delta \mathbf{k}_i^e X_i^e$$

Series Expansion Methods

$\hat{H}(x, \theta)$ can be exactly represented by an infinite series of random variables ξ_i and deterministic spatial functions φ_j

$$H(x, \theta) = \sum_{j=1}^{\infty} H_0 \varphi_j(x) + \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} H_{ij} \xi_{ij}(\theta) \varphi_j(x)$$

In practice $H(x, \theta)$ are truncated for M random variables ξ_i and N linearly independent deterministic spatial function $\varphi(x)$,

$$H(x, \theta) \approx H^{\text{series}}(x, \theta) = \sum_{j=1}^N H_0 \varphi_j(x) + \sum_{i=1}^M \sum_{j=1}^N H_{ij} \xi_{ij}(\theta) \varphi_j(x)$$

SEM: Properties of the RV

ξ_i are statistically independent random variables which satisfy the following properties:

$$\begin{cases} \mathbb{E}\{\xi_i\} = 0 & \forall i \in [1, M] \\ \mathbb{E}\{\xi_i \xi_j\} = \delta_{ij} & \forall (i, j) \in [1, M]^2 \end{cases}$$

with the Kronecker delta defined as ,

$$\delta_{ij} = \begin{cases} 1 & \forall i = j \\ 0 & \forall i \neq j \end{cases}$$

Series Expansion Methods

- Spectral Representation Method (SPRM)
- **Karhunen-Loève Expansion (KLE)**
- **Polynomial Chaos Expansion (PCE)**
- Expansion Optimal Linear Estimation Method (EOLE)
- Orthogonal Series Expansion (OSE)

Karhunen-Loève Expansion (KLE)

Any random field $H(x, \theta)$ with mean $\mu(x)$ and variance $\sigma(x)^2$ can be exactly expressed by an infinite series:

$$H(x, \theta) = \mu(x) + \sum_{i=0}^{+\infty} \sqrt{\lambda_i} f_i(x) \xi_i(\theta)$$

$f_i(x)$: orthogonal deterministic functions ,

$\xi_i(\theta)$: uncorrelated random variables with zero mean,

λ_i : set of constants (eigenvalues),

In practice

$$H(x, \theta) \approx \hat{H}^{\text{KLE}} = \mu(x) + \sum_{i=0}^M \sqrt{\lambda_i} f_i(x) \xi_i(\theta)$$

Karhunen-Loève Expansion (KLE)

The functions $f_i(x)$ are determined from the covariance function C_{HH} of the random field $H(x, \theta)$:

$$\begin{cases} \forall i = 1, \dots, M & \int_{\mathcal{D}} C_{HH}(x_1, x_2) f_i(x_2) d\mathcal{D}_{x_2} = \lambda_i f_i(x_1) \\ \int_{\mathcal{D}} f(x) \otimes f(x) d\mathcal{D} = I \end{cases}$$

→ the first expression is called FREDHOLM INTEGRAL EQUATION,

→ the second expression describes the orthonormality of the eigenfunctions.

The optimal basis $\{\mathbf{f}_i\} \Leftrightarrow (\lambda, \mathbf{f})$ are the eigenpairs of the problem.

Karhunen-Loève Expansion (KLE)

- λ_i is the i —th eigenvalue,
- f_i are the corresponding eigenfunctions.
- By definition the covariance matrix is bounded, symmetric and (semi-) definite:
 - all eigenvalues λ_i are real,
 - the set of eigenfunctions is a basis in $\mathcal{L}^2(\mathcal{D})$.

KLE: Properties of the RV

- ξ_i form a set of orthogonal random variables:

$$\mathbb{E}\{\xi_i \xi_j\} = \delta_{ij}$$

with respect to the inner product defined by:

$$(X_n, X_m) = \mathbb{E}\{X_n X_m\} = \int_{\Theta} X_n(x) X_m(x) d\mathcal{P}(x)$$

- $\xi_i(\theta) = \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{D}} H(x, \theta) f_i(x) d\mathcal{D}$

→ in the case that $H(x, \theta)$ is a Gaussian random field, each random variable ξ_i is Gaussian.

KLE: Properties of the Approximation

- Variance of the KLE approximated RF

$$\text{Var} \left[\hat{H}^{\text{KLE}}(x, \theta) \right] = \sum_{i=1}^M \lambda_i f_i^2(x)$$

- Covariance of the KLE approximated RF

$$\text{Cov} \left[\hat{H}^{\text{KLE}}(x_1, \theta), \hat{H}^{\text{KLE}}(x_2, \theta) \right] = \sum_{i=1}^M \lambda_i f_i(x_1) f_i(x_2)$$

- Variance of the approximation error

$$\text{Var} \left[H(x) - \hat{H}^{\text{KLE}}(x) \right] = \sigma^2(x) - \sum_{i=1}^M \lambda_i f_i^2(x)$$

Constant Variance

For constant variance it is

$$\bar{X}(\omega, z) = \mu_X(z) + \sigma_X \sum_{i=1}^T \sqrt{\lambda_i} \phi_i(z) \xi_i(\omega)$$

The approximation error is estimated as

$$\epsilon_{\sigma^2}(z) = \frac{\text{Var} \{X(\omega, z) - \bar{X}(\omega, z)\}}{\text{Var} \{X(\omega, z)\}} \quad \bar{\epsilon}_{\sigma^2} = \frac{1}{|\mathcal{D}|} \int_{\mathcal{D}} \epsilon_{\sigma^2}(z) \, dz$$

And the covariance can be expressed as

$$\bar{C}_X(z, z') = \sigma_X^2 \bar{\Gamma}_X(z, z') = \sigma_X^2 \sum_{i=1}^T \sqrt{\lambda_i} \phi_i(z) \phi_i(z')$$

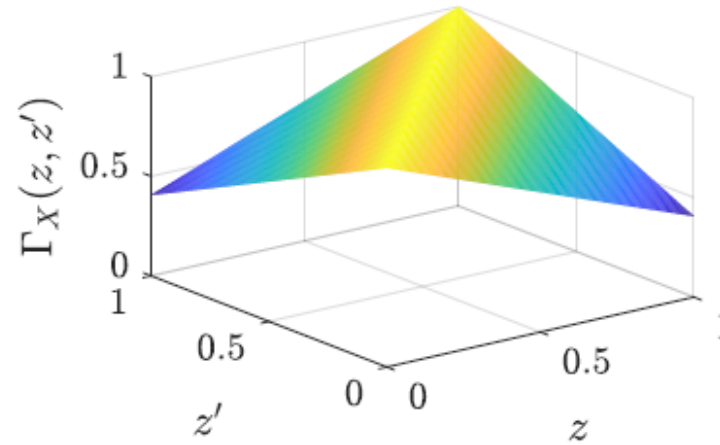
Correlation Function

kernel

correlation surface

correlation function

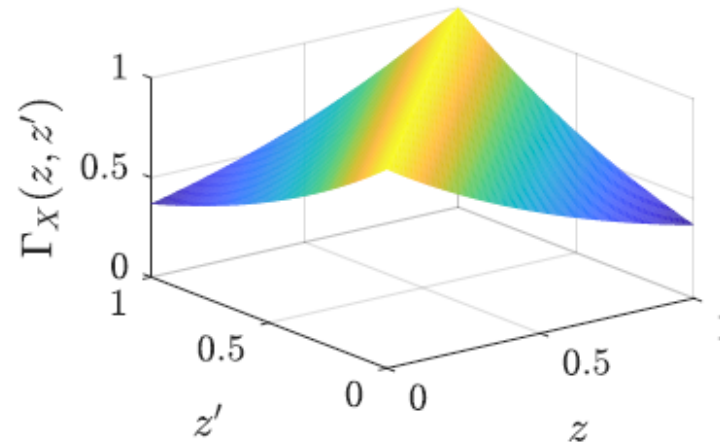
(a) binary
noise (BN)



$$\Gamma_X(z, z') = \max \left\{ 0 ; 1 - \frac{|z - z'|}{L_X^{\text{BN}}} \right\}$$

$$L_X^{\text{BN}} = 1.6835$$

(b) single
exponential (SE)



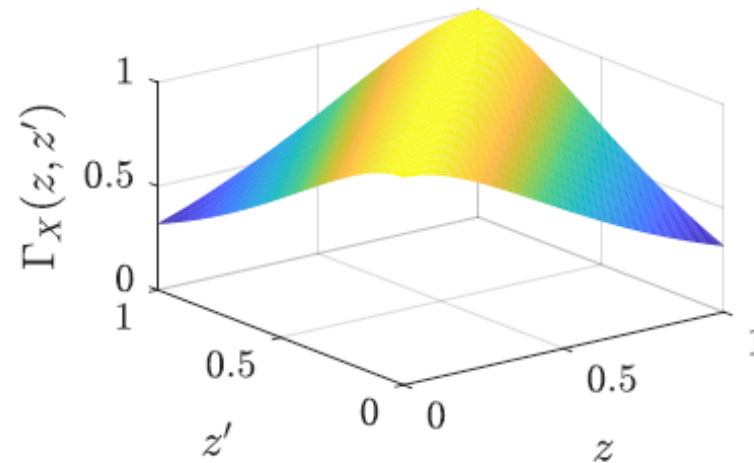
$$\Gamma_X(z, z') = \exp \left\{ -\frac{|z - z'|}{L_X^{\text{SE}}} \right\}$$

$$L_X^{\text{SE}} = 1.0$$

$$(\text{WM} : p = 0 \rightarrow \Gamma_X = \Gamma_{X, \text{WM}}^{0.5})$$

Correlation Function

(c) modified
exponential (ME)

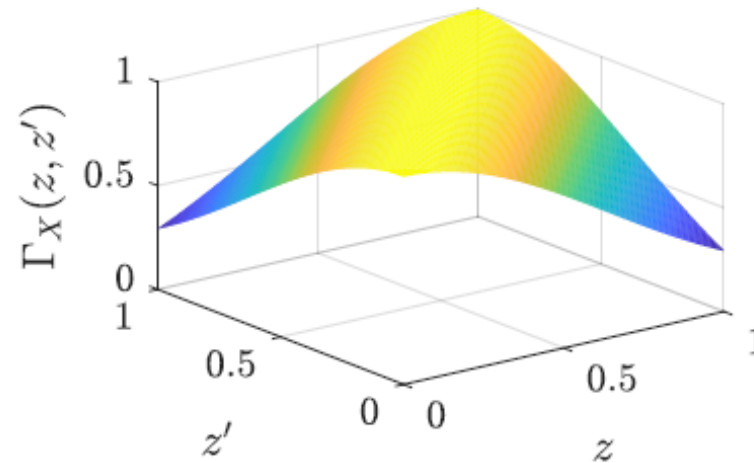


$$\Gamma_X(z, z') = \left(1 + \frac{|z - z'|}{L_X^{\text{ME}}}\right) \exp \left\{ -\frac{|z - z'|}{L_X^{\text{ME}}} \right\}$$

$$L_X^{\text{ME}} = 0.4249$$

$$(\text{WM} : p = 1 \rightarrow \Gamma_X = \Gamma_{X, \text{WM}}^{1.5})$$

(d) squared
exponential (QE)



$$\Gamma_X(z, z') = \exp \left\{ -\left(\frac{|z - z'|}{L_X^{\text{QE}}} \right)^2 \right\}$$

$$L_X^{\text{QE}} = 0.9018$$

$$(\text{WM} : p = \infty \rightarrow \Gamma_X = \Gamma_{X, \text{WM}}^{\infty})$$

Galerkin Solution of the Fredholm Integral Equation

- Fredholm Integral Equation:

$$\int_{\mathcal{D}} C_{HH}(x_1, x_2) f_i(x_2) d\mathcal{D}_{x_2} = \lambda_i f_i(x_1)$$

- Ansatz-functions for the eigenfunctions

$$f_k(x) = \sum_{i=1}^{\infty} d_i^k h_i(x)$$

- Error after truncation for N series terms

$$\epsilon_N(x) = \sum_{i=1}^N d_i^k \left[\int_{\mathcal{D}} C(x_1, x_2) h_i(x_2) d\mathcal{D}_{x_2} \lambda_i h_i(x_1) \right]$$

- Galerkin orthogonality of the residuum leads to

$$\sum_{i=1}^N d_i^k \left[\int_{\mathcal{D}} \left(\int_{\mathcal{D}} C(x_1, x_2) h_i(x_2) d\mathcal{D}_{x_2} \right) h_j(x_1) d\mathcal{D}_{x_1} - \lambda_k \int_{\mathcal{D}} h_i(x_1) h_j(x_1) d\mathcal{D}_{x_1} \right] = 0$$

Galerkin Solution of the Fredholm Integral Equation

we finally end up with a linear eigenvalue problem

$$\mathbf{W}\mathbf{D} = \mathbf{\Lambda}\mathbf{A}\mathbf{D}$$

with

$$A_{ij} = \int_{\mathcal{D}} h_i(x) h_j(x) d\mathcal{D}_{x_1}$$

$$W_{ij} = \int_{\mathcal{D}} \int_{\mathcal{D}} C(x_1, x_2) h_i(x) h_j(x) d\mathcal{D}_{x_1} d\mathcal{D}_{x_2}$$

$$D_{ij} = d_i^k \quad (\text{eigenvectors})$$

$$\Lambda_{ij} = \delta_{ij} \lambda_j \quad (\text{eigenvalues})$$

Galerkin Solution of the Fredholm Integral Equation

we finally end up with a linear eigenvalue problem

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with

$$A_{ij} = \int_{\mathcal{D}} h_i(x) h_j(x) d\mathcal{D}_{x_1}$$

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$$D_{ij} = d_i^k \quad (\text{eigenvectors})$$

$$\Lambda_{ij} = \delta_{ij} \lambda_j \quad (\text{eigenvalues})$$

Analytical Solution

For simple domains and certain correlation kernel an analytical solution can be given, e.g. exponential kernel or modified exponential kernel.

Exponential kernel:
$$\int_{-a}^a e^{-\frac{|\mathbf{x}_1 - \mathbf{x}_2|}{c_x}} f_i(\mathbf{x}_2) d\mathcal{D}_{\mathbf{x}_2} = \lambda_i f_i(\mathbf{x}_1)$$

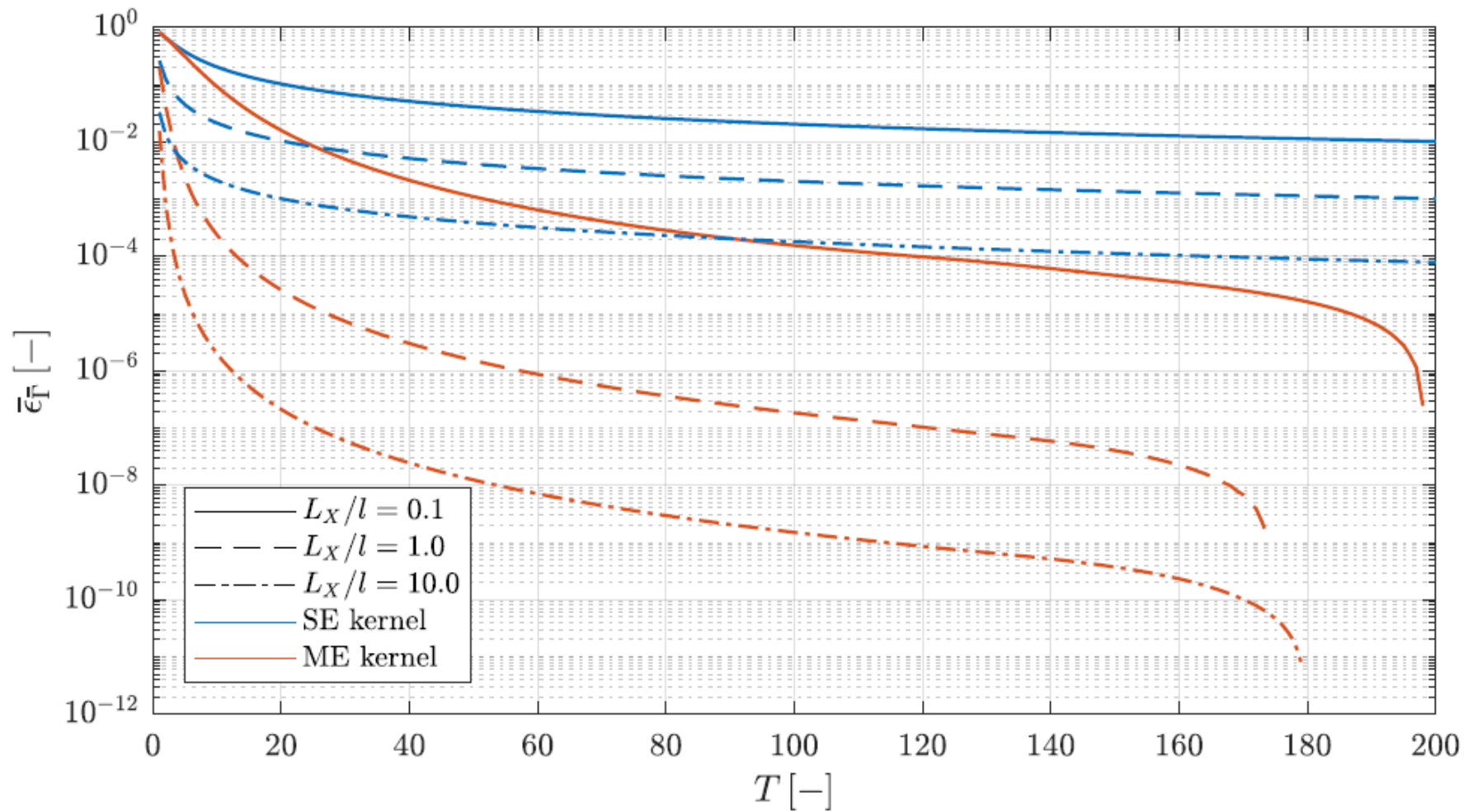
For $i = 1, 3, 5, \dots$
$$\lambda_i = \frac{2c}{\omega_i^2 c^2 + 1} \quad f_i(\mathbf{x}) = \alpha_i \cos(\omega_i \mathbf{x})$$

with
$$\alpha_i = \frac{1}{\sqrt{a + \frac{\sin(2\omega_i a)}{2\omega_i}}}$$

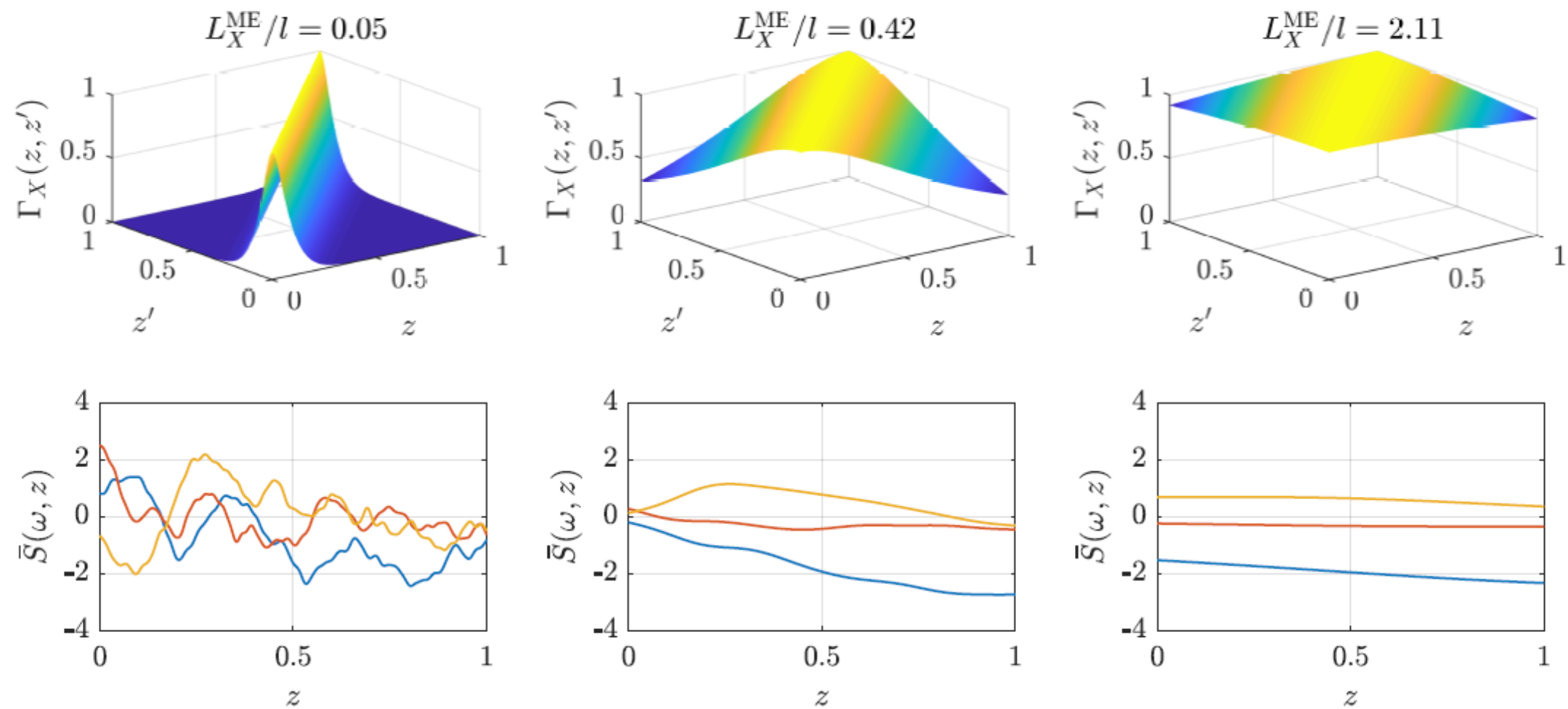
and for ω_i :
$$\frac{1}{c} - \omega_i \tan(\omega_i a) = 0 \quad \text{in the range} \quad \left[(i-1)\frac{\pi}{a}, (i-\frac{1}{2})\frac{\pi}{a} \right]$$

Similar for $i = 2, 4, 6, \dots$

Convergence Behavior



Meaning of the correlation-length



Modified Exponential Kernel

Steps for KLE

- A. Choice for a covariance function describing the random field
- B. Computation of the eigenpairs:
 - Analytical solutions are available for some covariance functions and simple geometries (e.g. rectangular domain)
 - Numerical solution with Galerkin approach by applying basis functions (e.g. piecewise polynomials, Legendre polynomials, wavelets etc.)

Polynomial Chaos Expansion (PCE)

Let X be a Gaussian RV and $\{\xi_i(\theta)\}_{i=1}^{\infty}$ a set of independent Gaussian RV's on Θ , then X can be represented by the expansion

$$\begin{aligned} X &= a_0 \Gamma_0 \\ &+ \sum_{i_1=1}^{\infty} a_{i_1} \Gamma_1(\xi_{i_1}(\theta)) \\ &+ \sum_{i_1=1}^{\infty} \sum_{i_2=1}^{i_1} a_{i_1 i_2} \Gamma_2(\xi_{i_1}(\theta), \xi_{i_2}(\theta)) \\ &+ \sum_{i_1=1}^{\infty} \sum_{i_2=1}^{i_1} \sum_{i_3=1}^{i_2} a_{i_1 i_2 i_3} \Gamma_3(\xi_{i_1}(\theta), \xi_{i_2}(\theta), \xi_{i_3}(\theta)) + \dots \end{aligned}$$

where Γ_p are chaos polynomials of order p and a_p are deterministic coefficients.

Chaos Decomposition

Wiener chaos decomposition is defined as

$$\mathcal{L}^2(\Theta, \mathcal{F}, \mathcal{P}) = \bigoplus_{i=0}^{+\infty} \Gamma_i$$

where $\bigoplus$ denotes the operator of orthogonal summation.

- The subspaces Γ_i are orthogonal to each other in $\mathcal{L}^2(\Theta, \mathcal{F}, \mathcal{P})$.
- The subspace Γ_p of $\mathcal{L}^2(\Theta, \mathcal{F}, \mathcal{P})$ is called the p -th homogeneous chaos.

Properties of the Polynomial Basis

- It is convenient to use a set of orthogonal polynomials with respect to the inner product defined by:

$$(X_n, X_m) = \mathbb{E}\{X_n X_m\} = \int_{\Theta} X_n(x) X_m(x) d\mathcal{P}(x)$$

- Polynomials of order $p > 0$ have an expectation equal to zero.

$$\mathbb{E}\{\Gamma_i\} = 0, \quad i > 0$$

- The mean value of the product of two polynomials with respect to the inner product is equal to zero:

$$\mathbb{E}\{\Gamma_i \Gamma_j\} = 0, \quad i \neq j$$

The PCE Representation

The PCE approximates the random process by a truncated series:

- involving M random variables ξ_i ,
- and the subspace Γ_p of PCE with a finite dimension P ,
i.e. polynomials up to the order p .

For example for $M = 2$:

$$\begin{aligned}
 H(x, \theta) \approx \hat{H}(x, \theta)_{M,p}^{\text{PCE}} &= a_0 \Gamma_0 + \sum_{i_1=1}^2 a_{i_1} \Gamma_1(\xi_{i_1}(\theta)) \\
 &+ \sum_{i_1=1}^2 \sum_{i_2=1}^{i_1} a_{i_1 i_2} \Gamma_2(\xi_{i_1}(\theta), \xi_{i_2}(\theta)) \\
 &+ \sum_{i_1=1}^2 \sum_{i_2=1}^{i_1} \sum_{i_3=1}^{i_2} a_{i_1 i_2 i_3} \Gamma_3(\xi_{i_1}(\theta), \xi_{i_2}(\theta), \xi_{i_3}(\theta)) + \dots
 \end{aligned}$$

The Optimal Choice of Polynomials

The optimal PCE basis has to be chosen depending on the distribution function of the RF (ASKEY SCHEME):

	PDF	Polynomials	Support range
Continuous	Gaussian	Hermite	$]-\infty; +\infty[$
	Gamma	Laguerre	$[0; +\infty[$
	Uniform	Legendre	$[a; b]$
	Beta	Jacobi	$[a; b]$
Discrete	Poisson	Charlier	$\{0, 1, 2, \dots\}$
	Binomial	Krawtchouk	$\{0, 1, 2, \dots, N\}$
	Negative binomial	Meixner	$\{0, 1, 2, \dots\}$
	Hypergeometric	Hahn	$\{0, 1, 2, \dots, N\}$

PCE for Gaussian Fields

- For Gaussian fields the Hermite Polynomials are optimal.
- A Gaussian field $H_{\text{Gauss}}(x, \theta)$ can exactly represented as:

$$H(x, \theta) = \sum_{j=0}^{\infty} a_j \Gamma_h(\{\xi_h(\theta)\}_{h=1}^{\infty})$$

where $\Gamma_0 = 1$ and $\Gamma_j(\{\xi_h(\theta)\}_{h=1}^{\infty})$ are Hermite polynomials of independent standard normal variables ξ_h on Θ .

1-Dimensional Hermite Polynomials

Γ can be generated directly from the definition of the Hermite polynomials

$$h_M(\xi) = (-1)^M \frac{\partial^M (e^{-\frac{1}{2}\xi^2})}{\partial \xi^M} e^{\frac{1}{2}\xi^2}$$

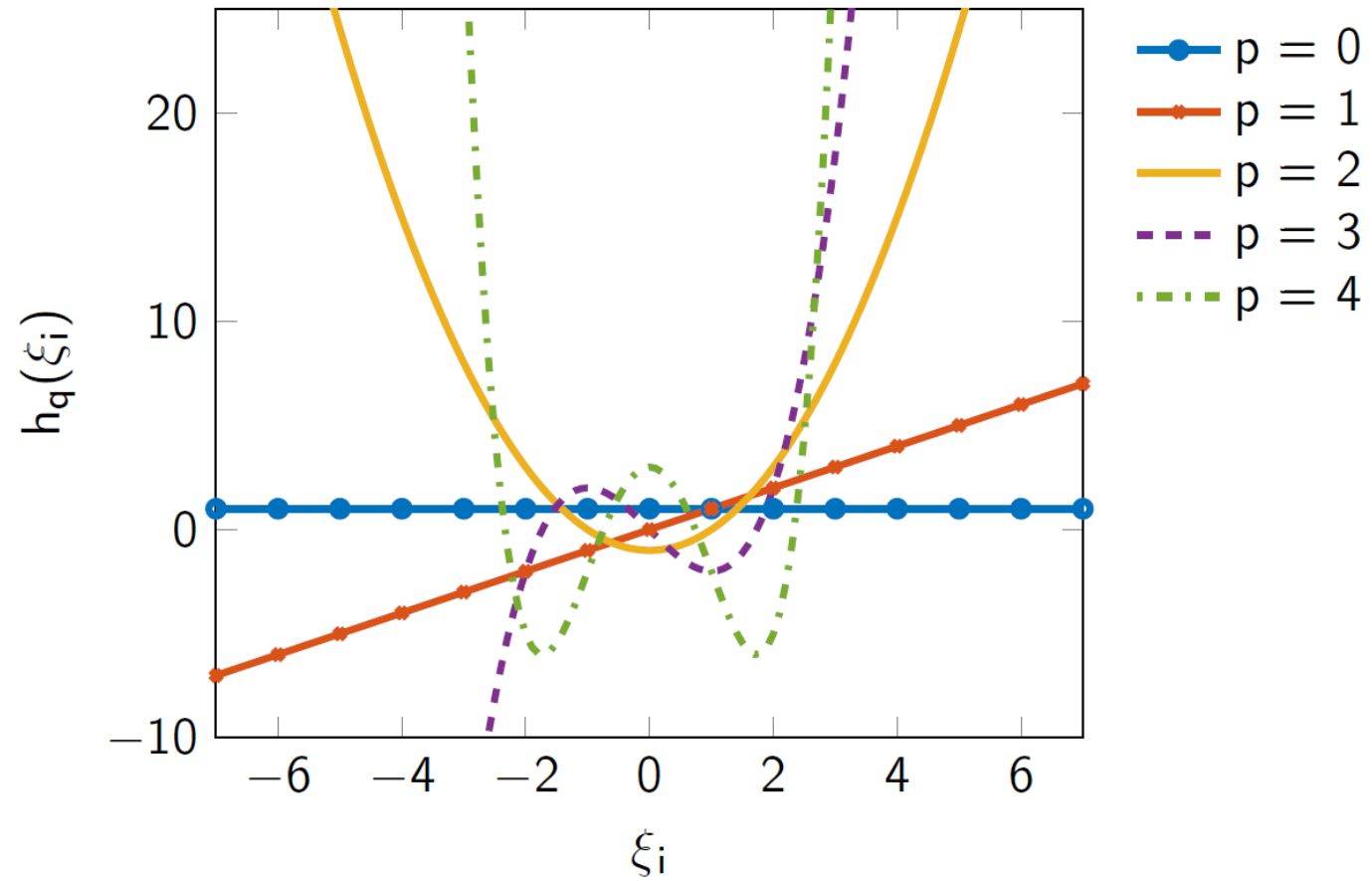
The first one-dimensional Hermite polynomials read

$$h_0 = 1, \quad h_1 = \xi, \quad h_2 = \xi^2 - 1, \quad h_3 = \xi^3 - 3\xi, \quad h_4 = \xi^4 - 6\xi^2 + 3$$

PCE approximation of the random field $H^{\text{Gauss}}(x, \theta)$:

$$\hat{H}_{\text{Gauss}}^{\text{PCE}}(x, \theta) = a_0 + a_1\xi + a_2(\xi^2 - 1) + a_3(\xi^3 - 3\xi) + a_4(\xi^4 - 6\xi^2 + 3)$$

1-Dimensional Hermite Polynomials



Dimension of the PCE

The number of polynomials (polynomial basis) needed for the dimension M (number of random variables) and polynomial order p is computed as:

$$\dim P = \frac{(M + p)!}{M!p!}$$

M	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
2	3	6	10	15	21	28
4	5	15	35	70	126	210
6	7	28	84	210	462	924

→ each scalar random value is represented by a series expansion with P coefficients

→ for larger fluctuations of the field higher order p of the polynomials gived better approximation.

Further developments PCE

- Generalized PCE [Xiu and Karniadakis (2002)]
- Adaptive multi-element gPC [Wan and Karniadakis (2006)]
- Adaptive sparse PCE [Blatman and Sudret (2010)]

Summary on RF Modeling

- Point discretization methods,
- Avaraging discretization methods,
- Series expansion methods:
 - KLE:
 - requires a priory knowledge on the correlation structure
 - cannot be used to approximate the systems response
 - PCE:
 - Can accurately describe any distribution
 - Does not require any a priory knowledge about the probability distribution